In [ ]:
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# PORTFOLIO #6: LIST
# CONCEPT: KINEMATIC DISPLACEMENT
```

In the topic of computational kinematics, potential energy represents the stored energy an object possesses due to its position in gravitational field.

In this notebook, we're going to explore classical physics and the integration of python programming. We're going to calculate the potential energy of an object at different heights.

In [10]:
```
# Serves as the dictionary, it is used to store data value pairs
# Allows for faster lookups, instead of searching for a list of items
heights = [0, 5, 10, 15, 20]

# Define constants for mass (kg) and gravity (m/s^2)
# Assigns a value to a variable
mass = 10
gravity = 9.81

# Create a new list for Potential Energy
energies = []

# Use a simple loop to calculate and add values to the list
for h in heights:
    # Executes the given formula
    pe = mass * gravity * h
    # Adds a single item to the end of an existing list
    energies.append(pe)

# Display the results simply
print("Height (m)  |  Energy (J)")
print("-" * 30)

for i in range(len(heights)):
    print(heights[i], "          |", energies[i])
```

```
Height (m)  |  Energy (J)
------------------------------
0           | 0.0
5           | 490.50000000000006
10           | 981.0000000000001
15           | 1471.5000000000002
20           | 1962.0000000000002
```